# October 16, 2026

## Recall from last time

Let $X_\Sigma$ be a complete toric variety with character lattice $M$, and let

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

be a torus-invariant Cartier divisor. Its polytope is

$$P_D = \{m \in M_\mathbb{R} : \langle m, u_\rho \rangle \geq -a_\rho \text{ for every } \rho \in \Sigma(1)\}.$$

The lattice points of $P_D$ index a basis of monomial global sections of $\mathcal{O}_{X_\Sigma}(D)$. When this set is nonempty, the complete linear system defines a rational map

$$\varphi_D : X_\Sigma \dashrightarrow \mathbb{P}^r, \qquad r = |P_D \cap M| - 1.$$

- $D$ is *basepoint free* if its global sections have no common zero. The section formula then defines $\varphi_D$ everywhere.
- $D$ is *ample* if $nD$ is very ample for some integer $n > 0$.
- $D$ is *very ample* if it is basepoint free and the induced morphism $\varphi_D$ is a closed embedding, identifying $X_\Sigma$ with its image in projective space.

# 1 Basepoint-free divisors on toric varieties

## 1.1 Sections as rational functions

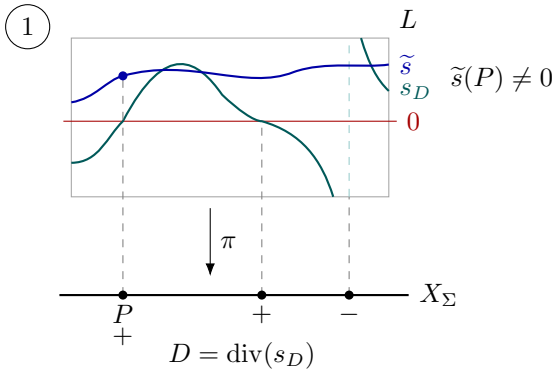
Put $L = \mathcal{O}_{X_\Sigma}(D)$. There is a distinguished rational section $s_D$ with $\operatorname{div}(s_D) = D$. It need not be regular when $D$ has negative coefficients. Every other nonzero rational section can be written uniquely as

$$\widetilde{s} = f s_D, \qquad f = \frac{\widetilde{s}}{s_D},$$

and

$$\operatorname{div}(\widetilde{s}) = \operatorname{div}(f) + D.$$

To test basepoint freeness at $P$, we ask whether there is a *global regular* section $\widetilde{s}$ with $\widetilde{s}(P) \neq 0$.




Schematic curve picture: positive coefficients record zeros of $s_D$, and negative coefficients record poles.

On a smooth curve, write $a_P = \mathrm{ord}_P(s_D)$. For a regular section,

$$\widetilde{s}(P) \neq 0 \quad \Longleftrightarrow \quad \mathrm{ord}_P(f) = -a_P.$$

Thus a zero of order $a_P > 0$ must be cancelled by a pole of that order in $f$, and a pole of $s_D$ must be cancelled by a zero of $f$. The orders need not be one. In higher dimensions, divisor orders are taken along prime divisors rather than at arbitrary points.

## 1.2 Torus eigensections and affine charts

The torus action lets us work with eigensections

$$s_m = \chi^m s_D, \qquad m \in M.$$

Since a character has zeros and poles only along the toric boundary,

$$\mathrm{div}(s_m) = \sum_{\rho \in \Sigma(1)} \big(\langle m, u_\rho \rangle + a_\rho\big) D_\rho.$$

Consequently,

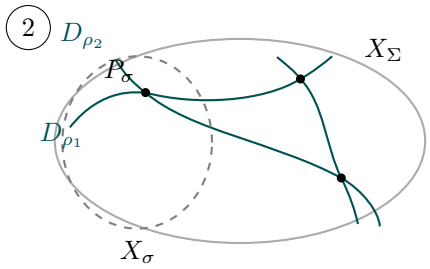
$$s_m \text{ is globally regular} \quad \Longleftrightarrow \quad m \in P_D \cap M.$$

Every global regular section is a linear combination of these eigensections. The rational function $\chi^m$ may have poles even when the section $s_m$ is regular: its poles are allowed to cancel the positive part of $D$.

Let $\sigma$ be a full-dimensional maximal cone, and write

$$X_\sigma = \mathrm{Spec}\, k[\sigma^\vee \cap M].$$

This affine chart has a distinguished torus-fixed point $P_\sigma$, the closed orbit corresponding to $\sigma$. Every invariant divisor $D_\rho$ with $\rho \in \sigma(1)$ passes through $P_\sigma$. These are exactly the invariant boundary divisors meeting the chart.




The toric boundary, its fixed points, and the affine chart $X_\sigma$ (schematic).

### 1.3 The section that does not vanish at $P_\sigma$

Because $D$ is Cartier, there is an integral vector $m_\sigma \in M$ such that

$$\langle m_\sigma, u_\rho \rangle = -a_\rho \qquad \text{for every } \rho \in \sigma(1).$$

The Cartier condition guarantees existence, even if the cone is not simplicial; full dimension guarantees uniqueness. The local equation and the corresponding local frame of $L$ are

$$g_\sigma = \chi^{-m_\sigma}, \qquad D|_{X_\sigma} = \operatorname{div}(g_\sigma), \qquad e_\sigma = g_\sigma^{-1} s_D = \chi^{m_\sigma} s_D.$$

The frame $e_\sigma$ is regular and nowhere zero on $X_\sigma$. An eigensection has local expression

$$s_m = \chi^{m-m_\sigma} e_\sigma.$$

If $s_m$ is regular on this chart, then $m - m_\sigma \in \sigma^\vee \cap M$. At $P_\sigma$, every nonconstant regular monomial vanishes. Since $\sigma$ is full dimensional,

$$\boxed{s_m(P_\sigma) \neq 0 \quad \Longleftrightarrow \quad m = m_\sigma.}$$

Equivalently, a regular eigensection is nonzero there precisely when

$$\langle m, u_\rho \rangle = -a_\rho \qquad \text{for all } \rho \in \sigma(1).$$

Such a section trivializes $L$ on the entire affine chart.

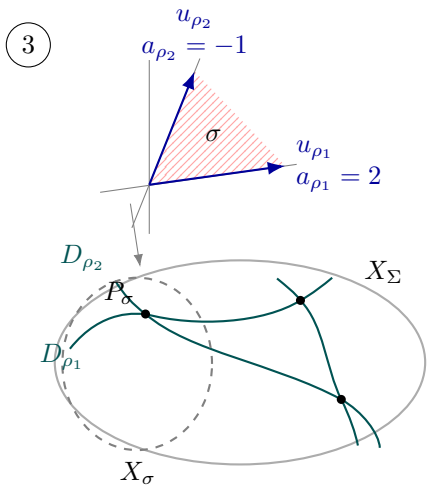
*Remark* 1. The eigensection hypothesis matters. A general regular section nonzero at the fixed point need not be nowhere zero on the chart: $1 + x$ on $\mathbb{A}^1$ is nonzero at 0 but vanishes at $-1$. Also, vanishing orders along the boundary alone do not detect nonvanishing of an arbitrary section at a fixed point: $x + y$ vanishes at the origin but has order zero along both coordinate axes.

### 1.4 The local model $x^2/y$

**Example 1.** Suppose $X_\sigma \cong \mathbb{A}^2$ is a smooth surface chart with $D_{\rho_1} = V(x)$ and $D_{\rho_2} = V(y)$. Take

$$D|_{X_\sigma} = 2V(x) - V(y) = \operatorname{div}\left(\frac{x^2}{y}\right).$$

Thus the distinguished rational section has a zero of order two along $V(x)$ and a pole of order one along $V(y)$.




The cone $\sigma$, its divisor coefficients, and its corresponding affine chart.

With respect to the dual basis determined by $x, y$,

$$m_\sigma = (-2, 1), \qquad e_\sigma = \frac{y}{x^2} s_D.$$

Write a regular section on the chart as $\widetilde{s} = p(x, y) e_\sigma$, where $p(x, y)$ is a polynomial. Its associated rational function is

$$f = \frac{\widetilde{s}}{s_D} = p(x, y) \frac{y}{x^2}.$$

We have $\widetilde{s}(P_\sigma) \neq 0$ exactly when $p(0, 0) \neq 0$. In that case,

$$\operatorname{ord}_{V(x)}(f) = -2, \qquad \operatorname{ord}_{V(y)}(f) = 1.$$

The eigensection corresponding to $p = 1$ is the local frame $e_\sigma = s_{m_\sigma}$. Its extension to a global section is determined by the remaining inequalities defining $P_D$.

## 1.5 The global basepoint-free criterion

For each maximal cone, the Cartier datum $m_\sigma$ gives the only possible exponent of an eigensection nonzero at $P_\sigma$. This candidate extends to a global regular section exactly when

$$m_\sigma \in P_D, \quad \text{that is,} \quad \langle m_\sigma, u_\rho \rangle \geq -a_\rho \quad \text{for every } \rho \in \Sigma(1).$$

The rays of $\sigma$ impose equalities; rays outside $\sigma$ give the additional tests for global regularity.

If a global regular section is nonzero at $P_\sigma$, one of its monomial summands must also be nonzero there. That summand has exponent $m_\sigma$, so $m_\sigma \in P_D$. Conversely, if $m_\sigma \in P_D$, the global section $s_{m_\sigma}$ is a frame on $X_\sigma$. Since these charts cover $X_\Sigma$,

$$\boxed{D \text{ is basepoint free} \quad \Longleftrightarrow \quad m_\sigma \in P_D \text{ for every maximal cone } \sigma.}$$

*Remark* 2 (The sign convention). Here $m_\sigma$ is the exponent of the local frame, so $\langle m_\sigma, u_\rho \rangle = -a_\rho$. If instead the Cartier data are written as exponents $q_\sigma$ of the local defining equations, then $q_\sigma = -m_\sigma$, and the same criterion reads $-q_\sigma \in P_D$. The minus sign comes from taking the reciprocal of the local defining equation to obtain a frame.

### 1.6 Reading regularity from the half-spaces

Each ray $\rho$ determines a half-space in the character space $M_{\mathbb{R}}$:

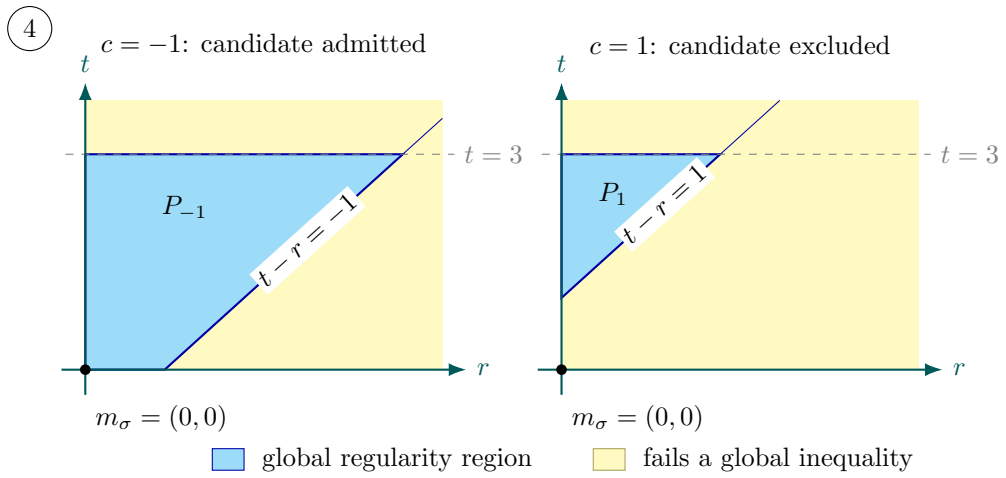
$$H_\rho = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -a_\rho\}, \qquad P_D = \bigcap_{\rho \in \Sigma(1)} H_\rho.$$

For a full-dimensional cone $\sigma$, the boundary hyperplanes associated to its rays meet at the Cartier datum $m_\sigma$. These equalities guarantee a local frame on $X_\sigma$. To extend that frame globally, $m_\sigma$ must also lie in every remaining half-space. Thus it is the section $s_{m_\sigma}$, rather than the lattice vector itself, whose global regularity we are testing.

The following model keeps the two local conditions $r, t \geq 0$ and their intersection $m_\sigma = (0, 0)$ fixed. The other two conditions are $t - r \geq c$ and $t \leq 3$, so

$$P_c = \{(r, t) : r \geq 0,\ t \geq 0,\ t - r \geq c,\ t \leq 3\}.$$

These inequalities come from the complete fan with rays $(1, 0)$, $(0, 1)$, $(-1, 1)$, and $(0, -1)$. Changing $c$ changes one divisor coefficient without changing $m_\sigma$.




Blue is the polytope whose lattice points index globally regular eigensections. Yellow lies in the local cone of candidates but outside the polytope. In the left panel, $m_\sigma \in P_{-1}$, so its section extends globally. In the right panel, $m_\sigma \notin P_1$, so it does not. The blue region is still nonempty there: other global sections exist, but all vanish at $P_\sigma$.

### 1.7 From Cartier data to a convex support function

To match the convention used on the board, write $q_\sigma = -m_\sigma$ for the exponent of the local defining equation. Define the piecewise-linear function $\psi_D : N_{\mathbb{R}} \to \mathbb{R}$ by

$$\psi_D(u) = \langle q_\sigma, u \rangle \quad \text{for } u \in \sigma.$$

The Cartier compatibility conditions make these linear functions agree on common faces, so $\psi_D$ is continuous. On the primitive ray generators,

$$\psi_D(u_\rho) = a_\rho.$$

The global regularity condition becomes

$$m_\sigma = -q_\sigma \in P_D \quad \Longleftrightarrow \quad \langle q_\sigma, u_\rho \rangle \leq a_\rho \quad \text{for every } \rho \in \Sigma(1).$$

Suppose these inequalities hold for every maximal cone $\sigma$. Fix one of its linear functions $\langle q_\sigma, \cdot \rangle$. On any other cone $\tau$, the ray inequalities and linearity give

$$\langle q_\sigma, u \rangle \leq \psi_D(u) \qquad (u \in \tau).$$

On $\sigma$ itself, equality holds. Conversely, expressing $\psi_D$ as this maximum gives the required inequalities at every ray. Hence

$$\boxed{D \text{ is basepoint free} \quad \Longleftrightarrow \quad \psi_D(u) = \max_{\sigma \in \Sigma(n)} \langle q_\sigma, u \rangle \,.}$$

In this convention, basepoint freeness is therefore equivalent to convexity of $\psi_D$. The maximum is taken over *linear functions*, evaluated at the same point $u$, rather than over lattice vectors. On a cone $\sigma$, its own linear function is an active maximum; all the other linear functions lie below it there.

*Remark* 3 (The opposite support-function convention). A common convention uses $\widehat{\psi}_D = -\psi_D$, with $\widehat{\psi}_D(u_\rho) = -a_\rho$. Then the same statement is

$$\widehat{\psi}_D(u) = \min_{\sigma \in \Sigma(n)} \langle m_\sigma, u \rangle \,.$$

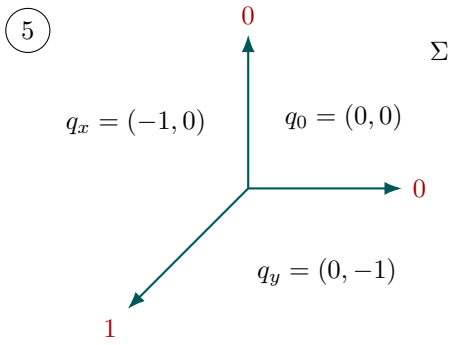
The maximum and minimum formulas express the same basepoint-free criterion in opposite sign conventions.

### 1.8 The example on $\mathbb{P}^2$

Consider the fan of $\mathbb{P}^2$ with primitive rays

$$u_1 = (1, 0), \qquad u_2 = (0, 1), \qquad u_3 = (-1, -1),$$

and divisor coefficients $a_1 = a_2 = 0$, $a_3 = 1$. The three local defining-equation exponents are $q_0 = (0, 0)$, $q_x = (-1, 0)$, and $q_y = (0, -1)$, on the corresponding cones shown below.




Ray coefficients are red; each cone is labelled by the exponent of its local defining equation.

The section exponents are the negatives of these vectors: $m_0 = (0,0)$, $m_x = (1,0)$, and $m_y = (0,1)$. All three lie in

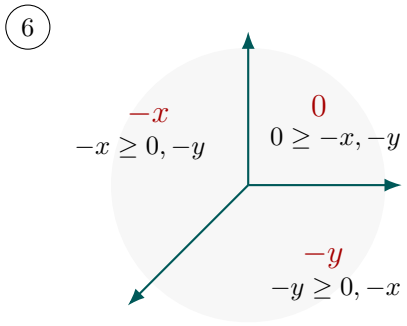
$$P_D = \{(r,t) : r \geq 0,\ t \geq 0,\ r + t \leq 1\},$$

so this divisor is basepoint free. In coordinates $u = (x,y) \in N_{\mathbb{R}}$, the support function is

$$\psi_D(x,y) = \begin{cases} 0, & x \geq 0,\ y \geq 0, \\ -x, & x \leq 0,\ y \geq x, \\ -y, & y \leq 0,\ x \geq y. \end{cases}$$

These pieces agree on the rays bounding adjacent cones. On each cone, the displayed piece is the largest of $0, -x, -y$. Therefore

$$\boxed{\psi_D(x,y) = \max(0, -x, -y).}$$




The active linear function on each cone.

## 2 Ample divisors and toric varieties from polytopes

With the support-function convention used above, the conclusion of the previous discussion is

$$\boxed{D \text{ is basepoint free} \iff \psi_D \text{ is convex.}}$$

We now turn to ampleness and very ampleness. This also answers Ayo's question: *If we start with a lattice polytope, is there a natural toric variety associated to it?*

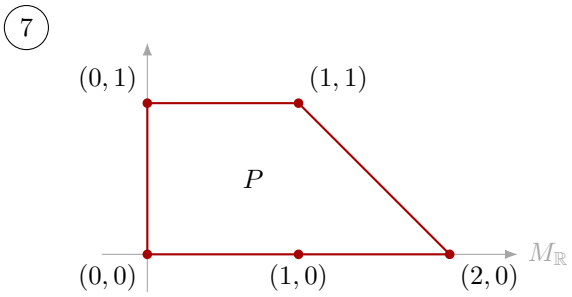
### 2.1 The trapezoid and its monomial map

Work over $\mathbb{C}$, and consider the trapezoid

$$P = \operatorname{conv}\{(0,0),(2,0),(1,1),(0,1)\} \subset M_{\mathbb{R}}.$$

Its five lattice points, in the order we will use for homogeneous coordinates, are

$$m_0 = (0,0), \quad m_1 = (1,0), \quad m_2 = (0,1), \quad m_3 = (2,0), \quad m_4 = (1,1).$$




The five lattice points of $P$ give the coordinates $1, s, t, s^2, st$.

The corresponding characters define a monomial map

$$\varphi_P : T = (\mathbb{C}^\times)^2 \longrightarrow \mathbb{P}^4, \qquad (s,t) \longmapsto [1 : s : t : s^2 : st].$$

Let

$$Y = \overline{\varphi_P(T)} \subseteq \mathbb{P}^4$$

be its Zariski closure. What variety is $Y$? We identify it by examining its affine charts.

Write $U_i = \{x_i \neq 0\} \subseteq \mathbb{P}^4$ and $Y_i = Y \cap U_i$. Since every monomial is nonzero on $T$, $\varphi_P(T)$ lies in every $U_i$, and $Y_i$ is its closure inside $U_i \cong \mathbb{A}^4$. In a normalized coordinate tuple, we list $x_j/x_i$ in increasing order of $j \neq i$.

## 2.2 The chart $U_0$: the vertex $(0,0)$

On $U_0$, use coordinates $\widetilde{x}_j = x_j/x_0$, for $j = 1, \ldots, 4$. The torus image is

$$(\widetilde{x}_1, \widetilde{x}_2, \widetilde{x}_3, \widetilde{x}_4) = (s, t, s^2, st).$$

Its closure is the graph of the polynomial map

$$\mathbb{A}^2 \longrightarrow \mathbb{A}^2, \qquad (s,t) \longmapsto (s^2, st).$$

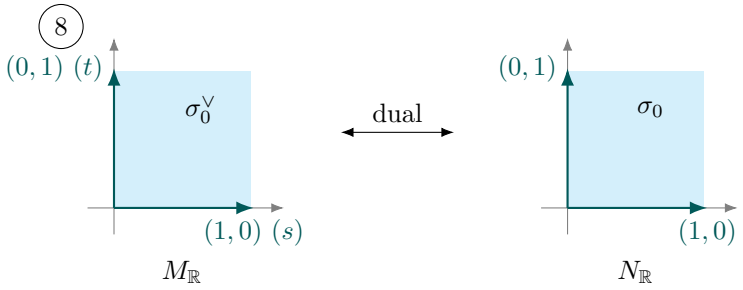
Equivalently,

$$Y_0 = V(\widetilde{x}_3 - \widetilde{x}_1^2, \widetilde{x}_4 - \widetilde{x}_1\widetilde{x}_2) \cong \mathbb{A}^2.$$

Its coordinate ring is $\mathbb{C}[s,t]$, so the character exponents generate

$$\sigma_0^\vee = \mathrm{cone}((1,0),(0,1)) \subset M_{\mathbb{R}}.$$

The dual cone $\sigma_0 \subset N_{\mathbb{R}}$ is also the first quadrant. Thus $Y_0 \cong X_{\sigma_0}$.




The chart associated with $(0,0)$ is $X_{\sigma_0} = \operatorname{Spec} \mathbb{C}[s,t] \cong \mathbb{A}^2$.

### 2.3 The chart $U_2$: the vertex $(0,1)$

Now normalize by $x_2$, and write $\widehat{x}_j = x_j/x_2$. The torus image in this chart is

$$(\widehat{x}_0, \widehat{x}_1, \widehat{x}_3, \widehat{x}_4) = \left(\frac{1}{t}, \frac{s}{t}, \frac{s^2}{t}, s\right).$$

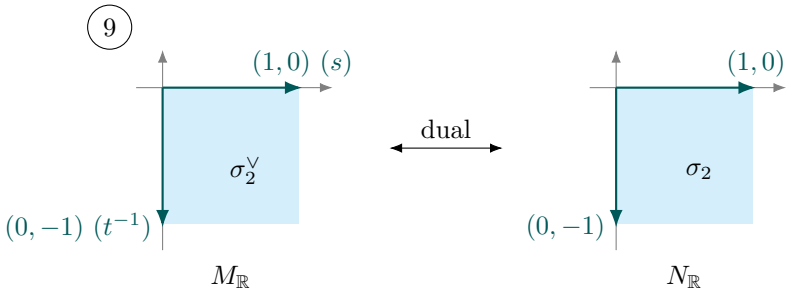
Set $u = s$ and $v = t^{-1}$. The tuple becomes

$$(v, uv, u^2v, u).$$

The first and last entries recover $v, u$, and the middle entries are polynomial functions of them. Its closure is therefore another graph isomorphic to $\mathbb{A}^2$, with coordinate ring $\mathbb{C}[s, t^{-1}]$. Hence

$$\sigma_2^\vee = \mathrm{cone}((1,0),(0,-1)), \qquad \sigma_2 = \mathrm{cone}((1,0),(0,-1)),$$

and $Y_2 \cong X_{\sigma_2}$ is the chart of the fourth-quadrant cone.




The chart associated with $(0,1)$ is $X_{\sigma_2} = \operatorname{Spec} \mathbb{C}[t^{-1}, s] \cong \mathbb{A}^2$.

### 2.4 The remaining vertex charts

The same calculation works at $(2,0)$ and $(1,1)$:

| chart and vertex | parameters | normalized image |
|---|---|---|
| $U_3,\ (2,0)$ | $u = s^{-1},\ v = t/s$ | $(u^2, u, uv, v)$ |
| $U_4,\ (1,1)$ | $u = s^{-1},\ v = s/t$ | $(u^2v, uv, u, v)$ |

In each row, two entries are $u, v$, and the others are polynomial functions of them. The closures $Y_3, Y_4$ are therefore both isomorphic to $\mathbb{A}^2$. Their cones are

$$\sigma_3^\vee = \mathrm{cone}((-1,0),(-1,1)), \quad \sigma_3 = \mathrm{cone}((0,1),(-1,-1)),$$
$$\sigma_4^\vee = \mathrm{cone}((-1,0),(1,-1)), \quad \sigma_4 = \mathrm{cone}((-1,-1),(0,-1)).$$

Thus $(2,0)$ gives the upper-left cone, and $(1,1)$ gives the narrow lower cone in the final fan.

The nonvertex lattice point $(1,0)$ gives a different kind of chart. On $U_1$, put $u = s$ and $v = t/s$. The coordinate tuple is

$$\left(\frac{1}{s}, \frac{t}{s}, s, t\right) = (u^{-1}, v, u, uv),$$

so

$$Y_1 \cong \operatorname{Spec} \mathbb{C}[u, u^{-1}, v] \cong \mathbb{C}^\times \times \mathbb{A}^1.$$

This chart corresponds to the ray $(0,1)$, rather than a maximal cone. It is not needed in the vertex cover: the relation $x_1^2 = x_0x_3$, which holds on $Y$, implies $Y_1 \subseteq Y_0 \cap Y_3$. In particular,

$$Y = Y_0 \cup Y_2 \cup Y_3 \cup Y_4.$$

## 2.5 Tangent cones and the normal fan

At a vertex $v$, normalizing the coordinate associated with $v$ replaces each monomial $\chi^m$ by $\chi^{m-v}$. These differences generate the tangent cone

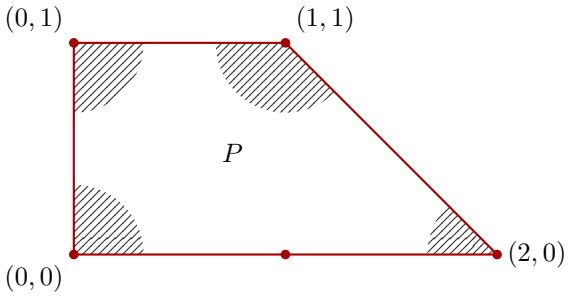
$$\sigma_v^\vee = \operatorname{cone}(P - v) \subset M_{\mathbb{R}}.$$

In this example they generate all its lattice points as a semigroup, as the two polynomial generators of each vertex chart show. Therefore the chart is exactly $X_{\sigma_v}$, where

$$\sigma_v = \{u \in N_{\mathbb{R}} : \langle m - v, u\rangle \geq 0 \text{ for all } m \in P\}.$$

Equivalently, $\sigma_v$ consists of linear functionals whose minimum on $P$ is attained at $v$. We use this *inward*, or minimizing, convention for normal cones.

10

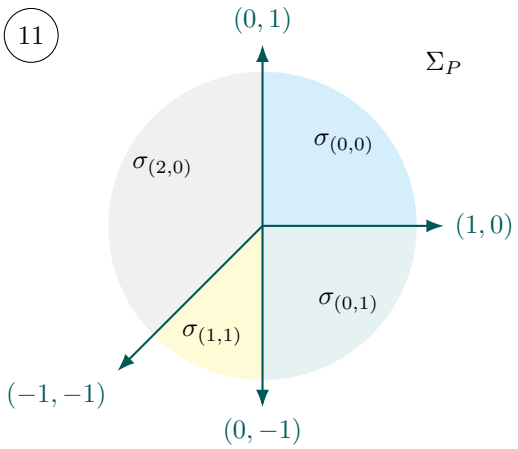



Black sectors: tangent cones $\sigma_v^\vee = \operatorname{cone}(P - v)$, translated back to vertices for display.

The four normal cones and all their faces form the normal fan $\Sigma_P$. Its rays are

$$(1, 0), \qquad (0, 1), \qquad (-1, -1), \qquad (0, -1).$$

11




The inward normal fan of $P$: each maximal cone consists of functionals minimized at the indicated vertex.

The affine charts glue along the monomial coordinate changes inherited from the same torus. We have therefore identified the projective closure:

$$\boxed{\overline{\varphi_P(T)} = Y \cong X_{\Sigma_P}.}$$

This answers Ayo's question: the normal fan is the combinatorial data of the natural normal toric variety associated to the polytope.

### 2.6 The connection with ampleness and very ampleness

The inequalities for our trapezoid are

$$r \geq 0, \qquad t \geq 0, \qquad t \leq 1, \qquad r + t \leq 2.$$

They define a torus-invariant Cartier divisor $D_P$ on $X_{\Sigma_P}$ with polytope $P$. In our support-function convention,

$$\psi_{D_P}(x, y) = \max(0, -y, -2x, -x - y).$$

Its maximal domains of linearity are exactly the maximal cones of $\Sigma_P$. This is *strict convexity relative to the fan*, the toric criterion for ampleness. It is stronger than the convexity criterion for basepoint freeness.

The explicit chart calculations give more in this example: the five monomial sections identify $X_{\Sigma_P}$ with the closed subvariety $Y \subseteq \mathbb{P}^4$. Hence $D_P$ is very ample.

*Remark* 4. A full-dimensional lattice polytope always determines a normal fan and a normal projective toric variety, together with an ample divisor. Identifying this variety with the direct closure of the lattice-point map requires an additional check in general; here the vertex-chart calculations supply it. For an ample divisor, some positive multiple is always very ample.